Steps to run the code:

1. For evaluation run the code in the file “Evaluation_ChainOfThoughtPrompt.ipynb” in the google colab environment: (note that the file is at https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/Evaluation_ChainOfThoughtPrompt.ipynb
    a. Input the file https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/aapl_trading_sentiment_data_all_days_RefPaper%20(1).csv

       This file contains the trade and sentiment data. Note that the sentiment signals were generated using the simple prompt used in the reference paper. The sentiment signals were generated using the DeepSeek-V3.
    b. Input the file https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/chainofthought_aapl_trading_sentiment_data_all_days.csv

       This file contains the trade and sentiment data. Note that the sentiment signals were generated using the ‘chain of thought’ prompt. The sentiment signals were generated using the DeepSeek-V3.

2. Any of the following model files can be used to regenerate the RL trading agents:

    a. https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/ppo_chainofthought_latest.zip
    b. https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/ppo_chainofthought_policy.pth
    c. https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/ppo_refpaper_latest.zip
    d. https://github.com/SatishChandraPhD/EvaluationFinRL2025/blob/main/EvaluationCodeWithModels/ppo_refpaper_policy.pth

Evaluation Results:

```
Rachev Ratio (α=0.05) for FewShot Agent: 0.9316
Rachev Ratio (α=0.05) for RefPaper Agent: 0.9042

=== Performance Comparison ===
           Metric   FewShot  RefPaper
     Total Return 0.078088  0.072370
    Annual Return 0.120755  0.113106
```

```
Annual Volatility 0.120535  0.123613
     Sharpe Ratio 1.001830  0.915002
     Max Drawdown 0.098987  0.101649
         Win Rate 0.395210  0.395210

Information Ratio: 0.0561
CVaR (95%): -0.0186
Outperformance Frequency: 55.95%
```

Analysis of the evaluation results:

Metric	**Value**	**Interpretation**
Information Ratio	0.0561	Small but positive edge over benchmark
CVaR (95%)	-0.0186	Moderate losses on worst 5% days
Outperformance Frequency	55.95%	Good — winning on most days